

UNIVERSITY OF TECHNOLOGY SYDNEY



32146 Data Visualisation and Visual Analytics - Autumn 2025

Assessment Task 2: Advanced Data Visualisation

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Australia - 2025

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I. Introduction

This report is based on the Australian International Trade dataset, which was extracted from official ABS (Australian Bureau of Statistics) sources. The dataset spans over three decades, from 1988 to 2024, and provides comprehensive trade data for both imports and exports.

The dataset comprises: 10 main trade categories, and 67 subcategories that reflect detailed trade performance across multiple industries and product groups.

These main categories represent a wide range of economic sectors and are classified as follows:

0. Food and live animals
1. Beverages and tobacco
2. Crude materials, inedible, except fuels
3. Mineral fuels, lubricants and related materials
4. Animal and vegetable oils, fats and waxes
5. Chemicals and related products, e.s.
6. Manufactured goods classified chiefly by material
7. Machinery and transport equipment
8. Miscellaneous manufactured articles
9. Commodities and transactions not classified elsewhere in the SITC

In this report, Category 3: Mineral fuels, lubricants and related materials is selected for detailed analysis, with a particular focus on Subcategory 33: *Petroleum, petroleum products and related materials*.

II. Data preparation

Effective data preparation is essential for ensuring meaningful and accurate analysis, particularly when working with large-scale raw datasets. To support the identification of long-term trends and comparisons between trade categories, the original data was transformed into two additional pattern sheets: Statistical Pattern and Analytical Pattern.

2.1 Statistical Pattern

The Statistical Pattern calculates the relative proportion of each main category and its subcategories in relation to total import or export values for each year. This enables comparison of trade structures over time. The formula used is:

$$\text{Percentage (Import/Export)} = \frac{\text{Category (Subcategory) (Import/Export)}}{\text{Total (Import/Export)}} \times 100\%$$

This proportion is calculated separately for imports and exports on a yearly basis.

2.2 Analytical Pattern

The Analytical Pattern captures the year-over-year percentage change in trade values for each category. It is used to detect significant shifts, trends, and volatility in import/export activity. The first year is standardized to 100%, and subsequent years are compared to the previous one using the formula:

$$\text{Ratio Change Yearly} = \frac{d_y}{d_{y-1}} \times 100\%$$

Where:

- d_y is the trade value in year y
- d_{y-1} is the trade value in the previous year

This approach helps identify which categories have undergone the most substantial changes and provides insight into emerging trends across the trade landscape.

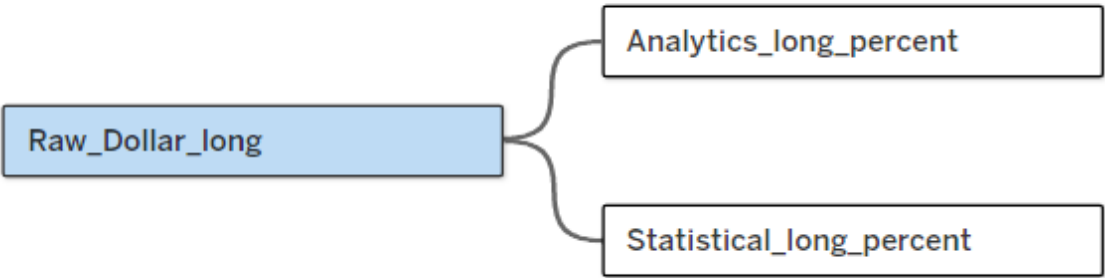
2.3 Data Reshaping for Tableau

To facilitate effective time-series analysis and ensure compatibility with Tableau's data structure, the original datasets were restructured from wide format to long format using Power Query in Microsoft Excel. This transformation was applied to the raw dollar values, statistical proportions, and analytical change patterns, resulting in three new sheets: *Raw_Dollar_Long*, *Statistical_Long_Percent*, and *Analytics_Long_Percent*. By converting multiple year-based columns into a single "year" variable with corresponding values, the data became more structured and flexible for filtering, aggregation, and visualization. This step was essential in supporting consistent trend analysis across time and categories, and in enabling the development of interactive dashboards within Tableau.

2.4 Tableau connectivity

The relationship between the three datasets was established in Tableau by connecting *Analytics_Long_Percent* and *Statistical_Long_Percent* to the central *Raw_Dollar_Long*

table. This connection enables integrated analysis across absolute values, percentage shares, and year-over-year changes within a unified data model, supporting comprehensive and comparative data visualizations.



III. Data Visualization

3.1. All main categories analysis

3.1.1. Raw dollar value trends

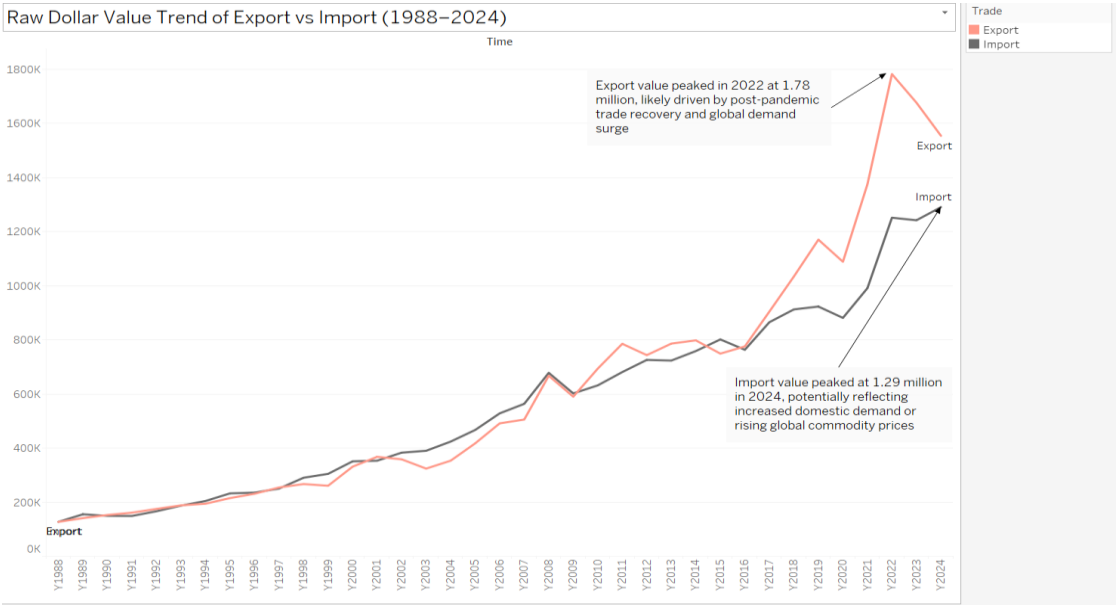


Figure 1 Raw Dollar value trends of Export and Import (1988-2024)

Figure 1 illustrates the raw dollar value trend of Australia’s total exports and imports from 1988 to 2024. Overall, both exports and imports have shown a steady upward trajectory, with notable acceleration after 2019. Export values peaked in 2022 at

approximately \$1.78 million, largely attributed to the global trade rebound following the COVID-19 pandemic, as reported by the United Nations Conference on Trade and Development (UNCTAD, 2022). Meanwhile, imports continued to rise and reached a peak of \$1.29 million in 2024, potentially reflecting increased domestic consumption and higher global commodity prices during that period (World Bank, 2022). The widening gap between export and import values in recent years suggests a growing trade surplus during the post-COVID recovery period.

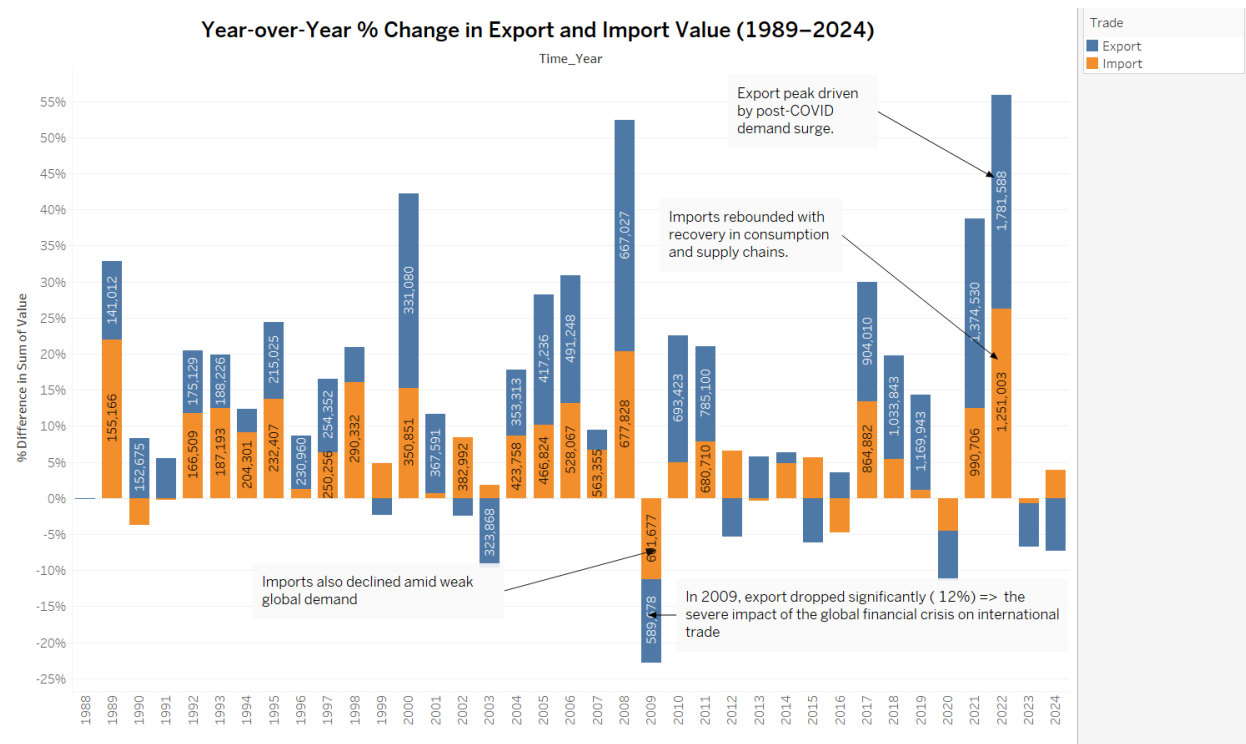


Figure 2: Year-over-Year % Change in Export and Import Value (1989–2024)

Figure 2 presents the year-over-year percentage change in Australia’s export and import values from 1989 to 2024. The chart reveals several notable periods of volatility. In 2009, exports experienced a sharp decline of approximately 12%, reflecting the severe impact of the global financial crisis on international trade. In contrast, 2022 saw the largest surge in export growth, exceeding 50%, largely due to the post-COVID recovery in global demand. Import growth also rebounded significantly in 2023–2024, coinciding with the normalization of supply chains and rising domestic consumption. The chart highlights that while export growth has historically been more volatile, major global events tend to drive synchronized fluctuations in both trade flows.

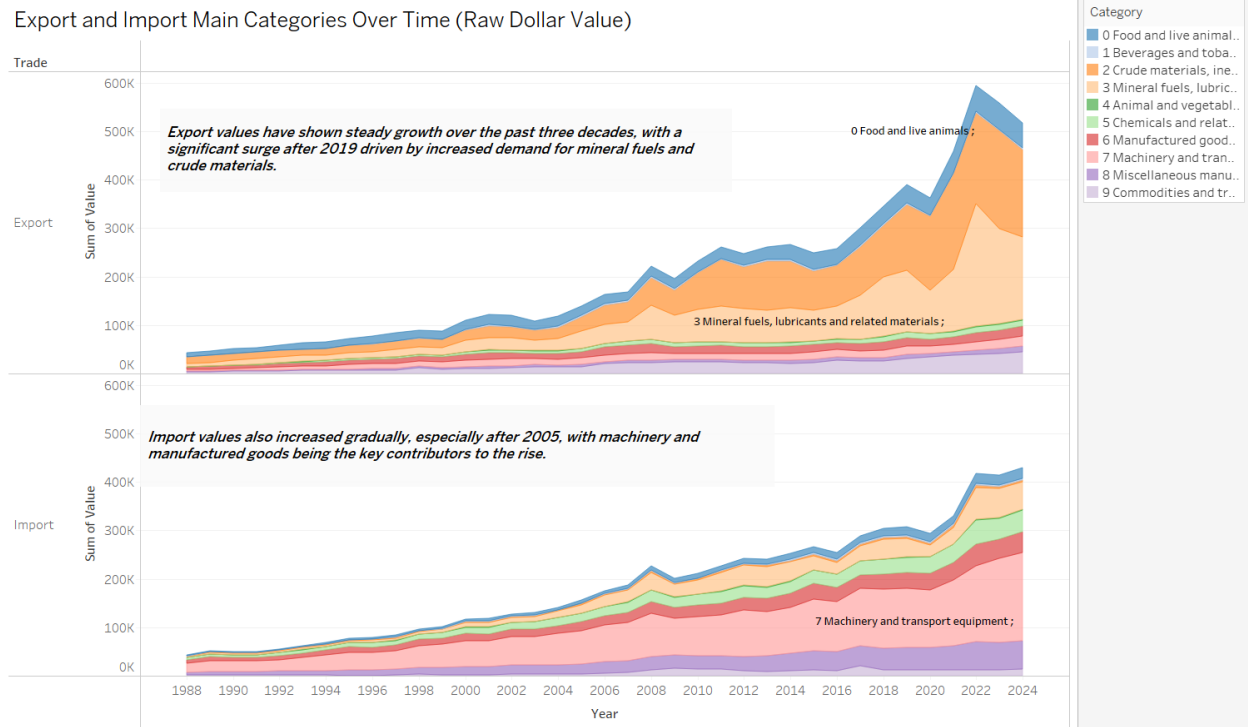


Figure 3 Export and Import Main Categories Over Time (Raw Dollar Value)

Figure 3 illustrates the raw dollar value trends of Australia's main export and import categories from 1988 to 2024. Export values have demonstrated consistent growth over the past three decades, with a significant surge after 2019 largely driven by increased demand for mineral fuels and food products. On the import side, a steady upward trend is also observed, especially after 2005, with machinery and transport equipment emerging as the dominant import category. These patterns suggest structural differences in Australia's trade profile, with resource-based exports and manufactured imports shaping long-term trade dynamics.

3.1.2. Analytical Pattern

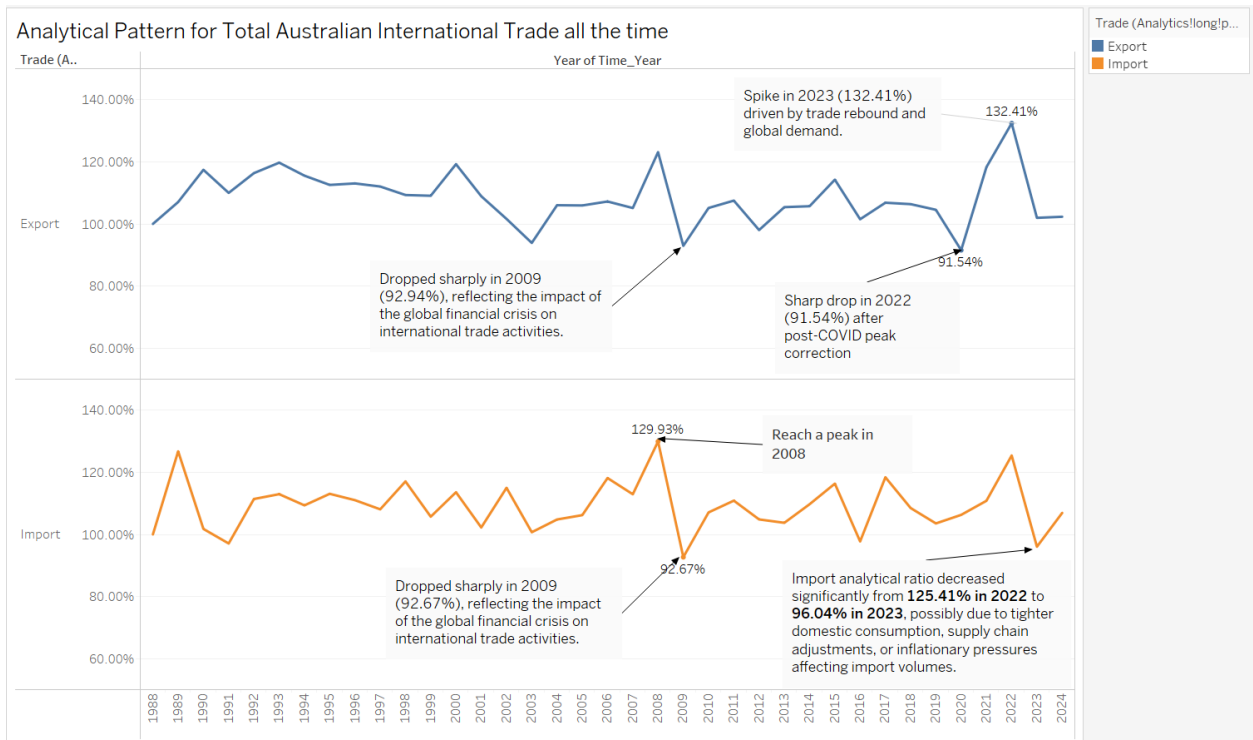


Figure 4 Analytical Pattern for Total Australian International Trade all the time

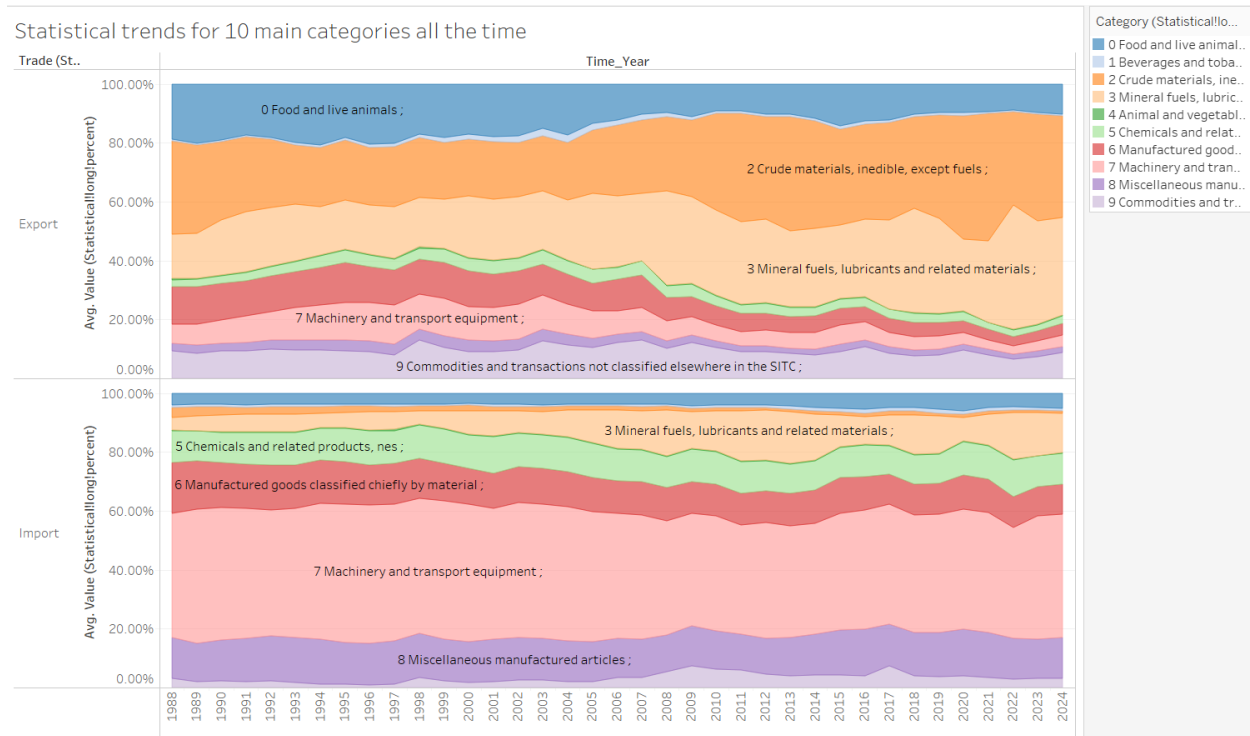
Figure 4 illustrates the analytical trends of Australia's international trade from 1988 to 2024. The export ratio experienced a notable drop in 2009 (92.94%) due to the global financial crisis, followed by a recovery phase with fluctuations. A sharp decline occurred again in 2022 (91.54%) after the post-COVID export surge, but rebounded strongly in 2023, reaching 132.41%. For imports, the analytical ratio also dipped significantly in 2009 (92.67%) and peaked earlier in 2008 (129.93%). A recent sharp decline is observed from 125.41% in 2022 to 96.04% in 2023, possibly linked to inflationary pressure, domestic consumption adjustments, or supply chain constraints.

Proportion of 10 main categories

Trade (St..



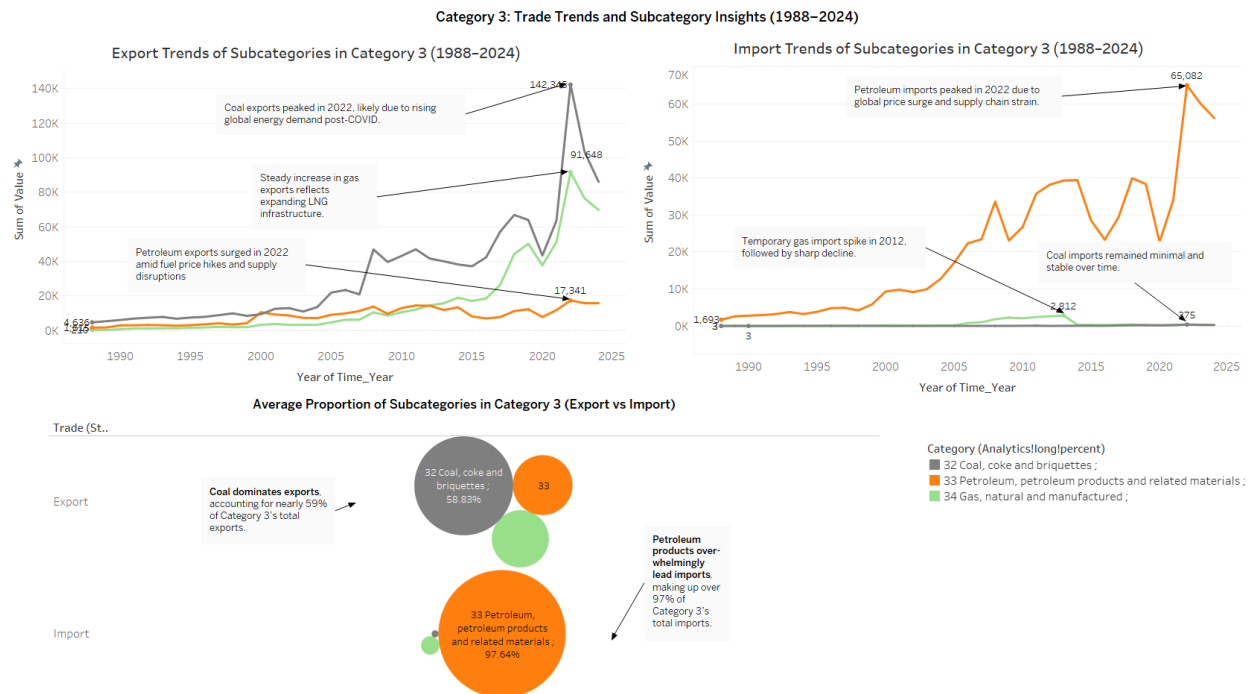
The bubble chart illustrates the overall proportion of Australia's main trade categories. In exports, Mineral fuels (Category 3) and Crude materials (Category 2) dominate the chart, highlighting their strong contribution to Australia's export economy. Food and live animals (Category 0) also show a significant export share. Conversely, imports are heavily concentrated in Machinery and transport equipment (Category 7), accounting for approximately 42.49% of total imports, followed by Miscellaneous manufactured articles (Category 8) and Chemicals (Category 5). This visualization emphasizes the country's export strength in natural resources and its reliance on industrial and technological imports.



The statistical trend chart shows the evolving proportional contributions of Australia's ten main export and imports categories over time. In exports, Crude materials (Category 2) and Mineral fuels (Category 3) consistently dominate, with Food and live animals (Category 0) increasing their share notably after 2010. In contrast, imports are heavily concentrated in Machinery and transport equipment (Category 7) throughout the years, maintaining the highest proportion, followed by Miscellaneous manufactured articles (Category 8) and Chemicals (Category 5). This visualization highlights Australia's long-standing export strengths in natural resources and its dependency on manufactured goods and machinery for imports.

3.2. Dashboard

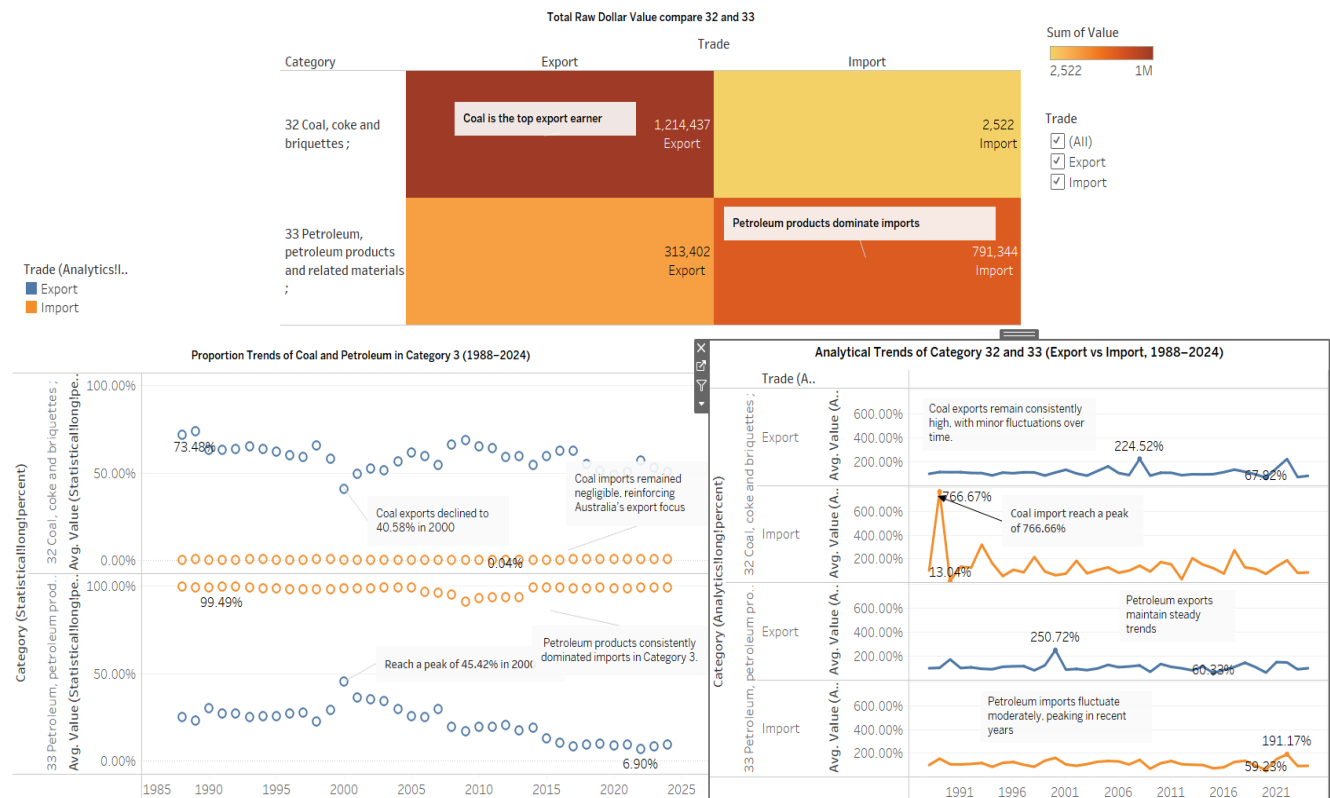
3.2.1. Category 3: Trade Trends and Subcategory Insights (1988–2024)



The dashboard illustrates a clear upward trend in exports within Category 3 (Coal, Petroleum, and Gas), especially for coal (Subcategory 32) and liquefied natural gas (Subcategory 34). Coal exports peaked in 2022, accounting for nearly 59% of Category 3's total exports. This trend reflects strong energy demand from the Asia-Pacific region and global supply disruptions, particularly from Russia, as reported by the IEA (2022). Similarly, LNG exports have consistently increased since 2010, supported by Australia's large-scale infrastructure development in LNG projects (Department of Industry, 2021). On the import side, Category 3 is dominated by petroleum and related products (Subcategory 33), which make up over 97% of total imports in this category. The year 2022 also saw a significant surge in oil imports, coinciding with a sharp increase in global crude oil prices, reaching nearly USD 100 per barrel according to Statista (2023). Overall, the data clearly demonstrate the correlation between global energy market fluctuations and Australia's trade performance across each subcategory within Category.

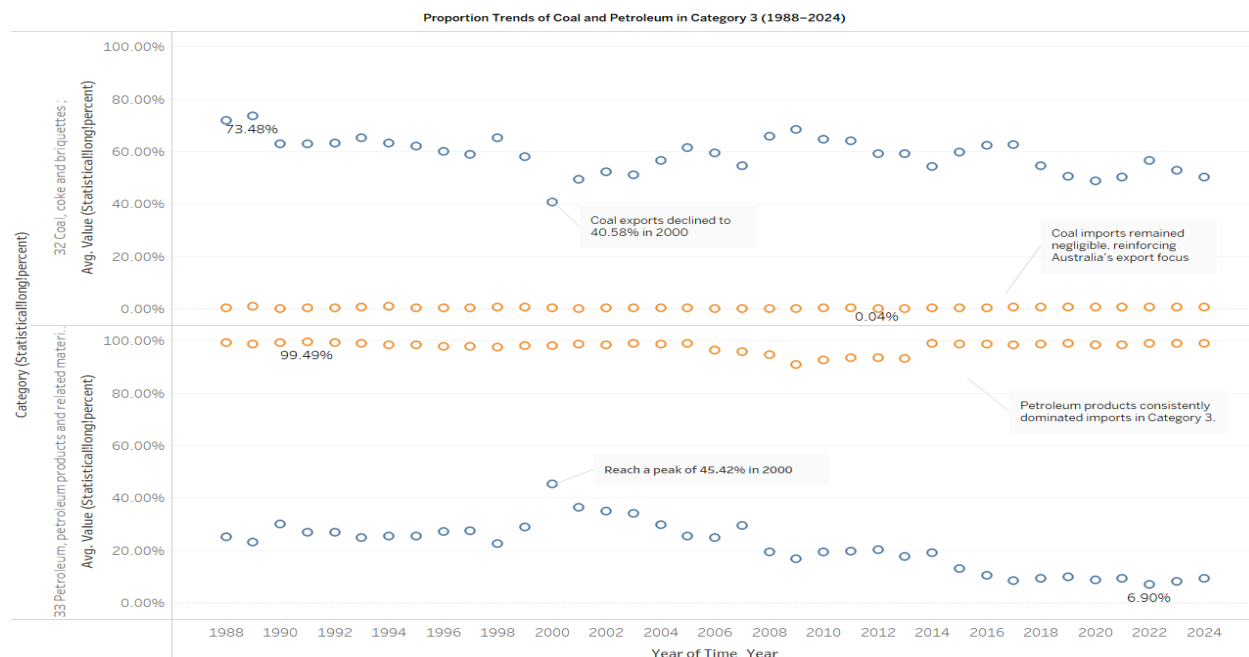
3.2.2. Comparative Analysis of Coal and Petroleum in Category 3

Comparative Trade Analysis of Coal and Petroleum (1988–2024)

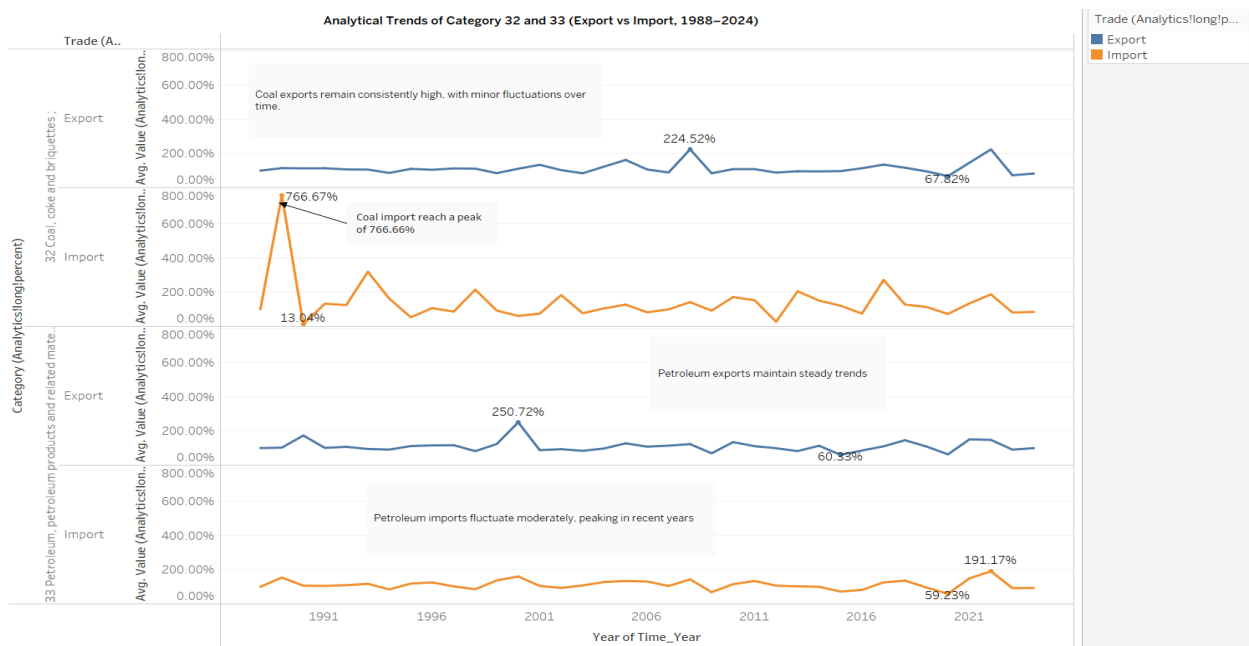


This dashboard presents a comparative analysis of Australia’s international trade performance between two major subcategories in Category 3: 32 (Coal, coke and briquettes) and 33 (Petroleum, petroleum products and related materials), using raw dollar values, statistical proportions, and analytical growth trends.

Based on the total raw dollar values, coal dominates export activities, contributing over 1.2 million, making it the leading subcategory in Australia’s energy exports. In contrast, petroleum products dominate imports with a value of over 791,000, reflecting Australia’s limited domestic oil refining capacity and reliance on imported fuels (Department of Climate Change, Energy, the Environment and Water, 2023).



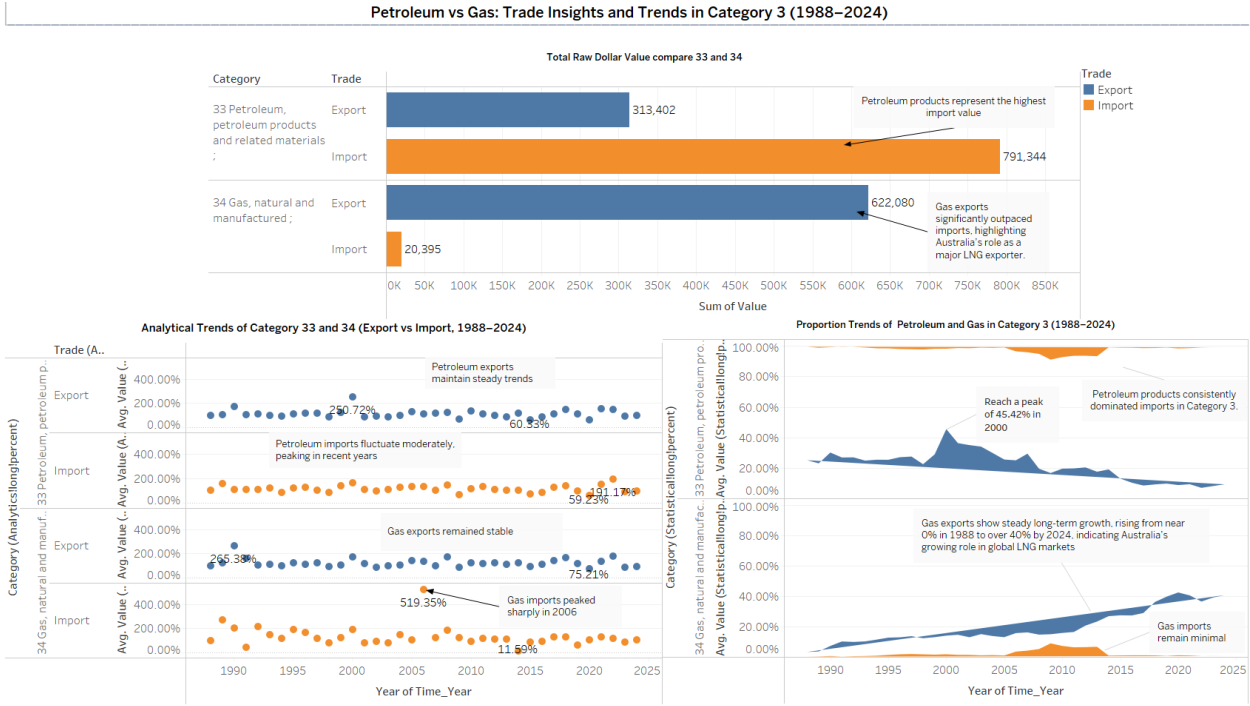
The statistical trend chart shows that coal consistently accounted for the largest proportion of exports, although it experienced a notable decline from 73.48% in 1988 to 40.58% in 2000. However, it stabilized again in the following decades. Meanwhile, petroleum maintained a dominant import share of nearly 100%, illustrating the strategic importance of petroleum-related goods to Australia's industrial and transport sectors.



The analytical trend chart highlights that coal exports maintained high but relatively stable growth, peaking at 224.52%, while coal imports spiked unexpectedly in the early 1990s

(766.67%), possibly due to temporary demand or international supply disruptions. On the other hand, petroleum exports reached a peak of 250.72%, likely reflecting price volatility and opportunistic overseas supply strategies. Petroleum imports, however, fluctuated moderately, peaking at 191.17% in recent years, which aligns with global oil price shocks during events like the COVID-19 recovery period and 2022 energy crisis (International Energy Agency, 2022; Statista, 2023).

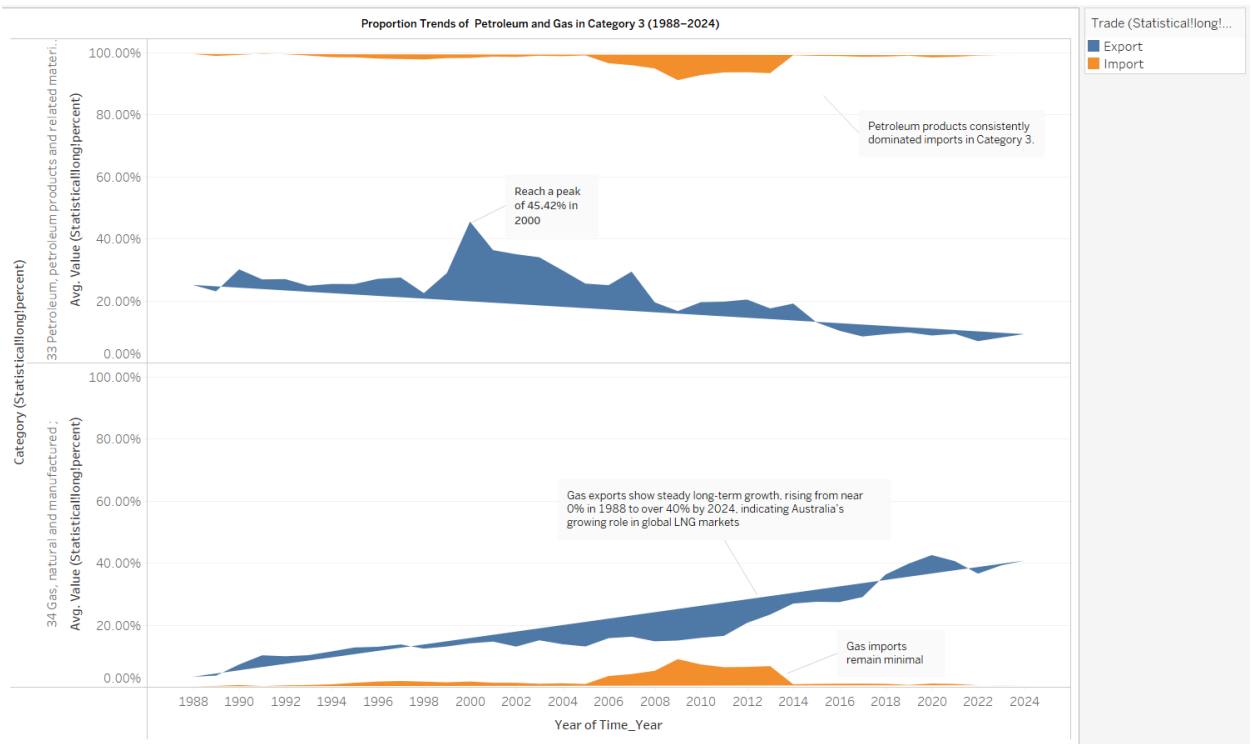
3.2.3. Petroleum vs Gas: Trade Insights and Trends in Category 3



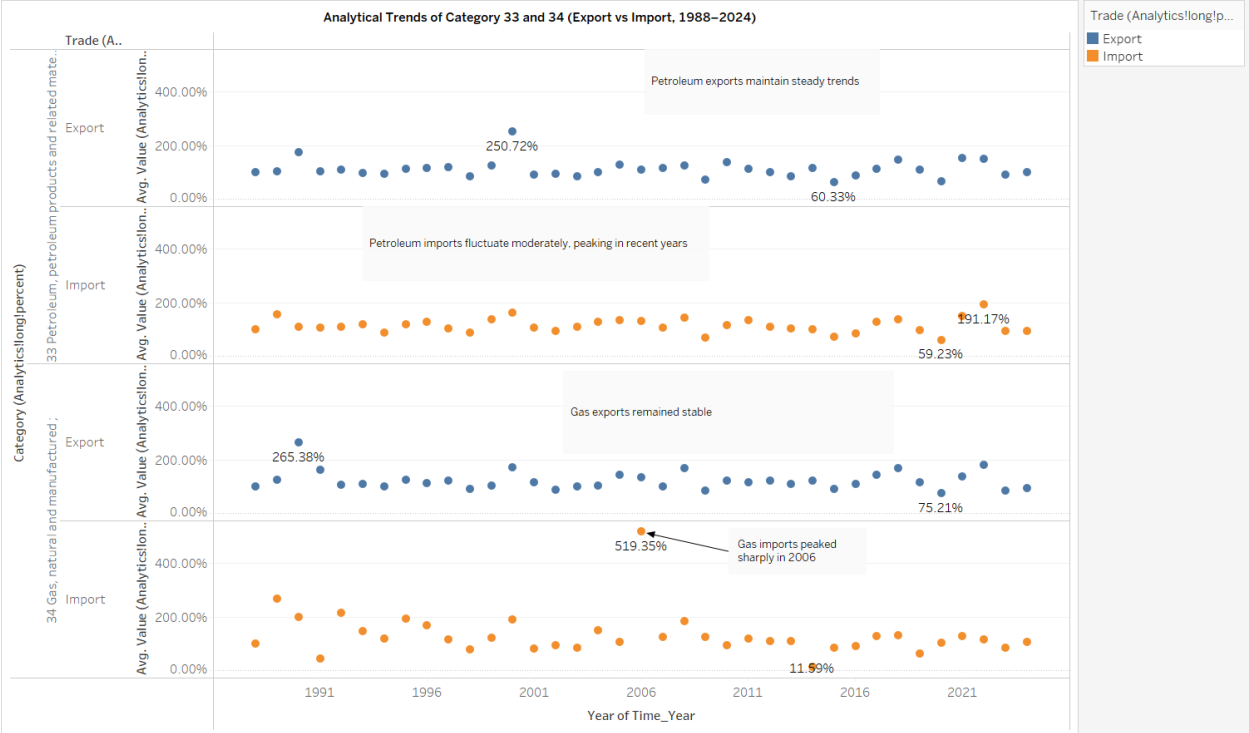
This dashboard offers a comprehensive examination of Australia's trade dynamics concerning two pivotal energy commodities: Petroleum (Category 33) and Gas (Category 34), spanning from 1988 to 2024. It integrates raw dollar values, analytical growth trends, and proportional contributions within Category 3 to elucidate the contrasting roles of these subcategories in Australia's trade landscape.

The raw dollar value analysis reveals that petroleum products dominate imports, with cumulative import values significantly surpassing exports. This trend underscores Australia's reliance on imported refined fuels, a consequence of the gradual closure of domestic refineries over recent decades. As of 2021, only two refineries remained operational, leading to increased dependence on imports to meet domestic fuel demand.

Conversely, gas exports have experienced substantial growth, positioning Australia as a leading exporter of liquefied natural gas (LNG). Major projects like the Gorgon Gas Project and the North West Shelf Project have been instrumental in this expansion, enabling Australia to supply LNG to key Asian markets.



The proportional analysis within Category 3 indicates that petroleum consistently constitutes most imports, reflecting its critical role in Australia's energy consumption. In contrast, gas has emerged as a dominant export, with its share increasing markedly over the period analyzed. This shift aligns with global energy trends and Australia's strategic investments in LNG infrastructure.



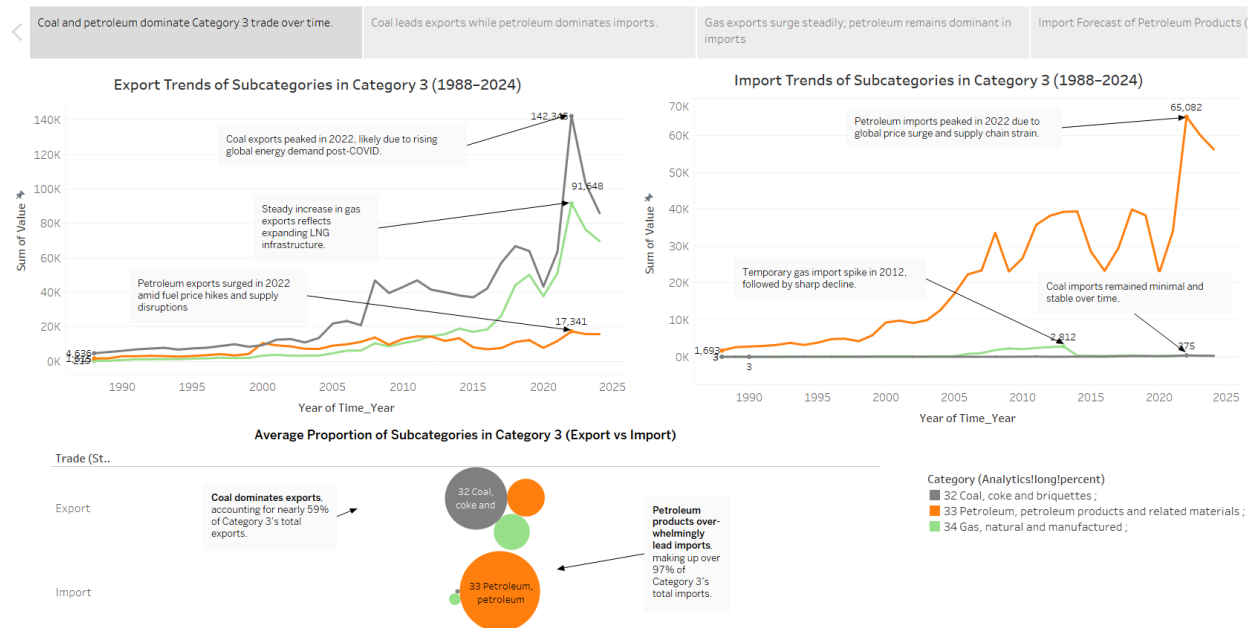
The analytical trends highlight steady growth in gas exports, driven by escalating global demand and Australia's capacity to fulfill long-term supply contracts. However, fluctuations are evident, influenced by factors such as global market volatility and geopolitical events. For instance, the 2022 global energy crisis, exacerbated by geopolitical tensions, led to heightened LNG demand and price surges.

Petroleum imports, while substantial, exhibit relative stability, with minor variations corresponding to global oil price movements and domestic consumption patterns.

3.3. Storyboard

3.3.1. Story 1 Analysis: Coal and petroleum dominate Category 3 (1988-2024)

Energy Trade Focus: Petroleum Products (1988–2029)



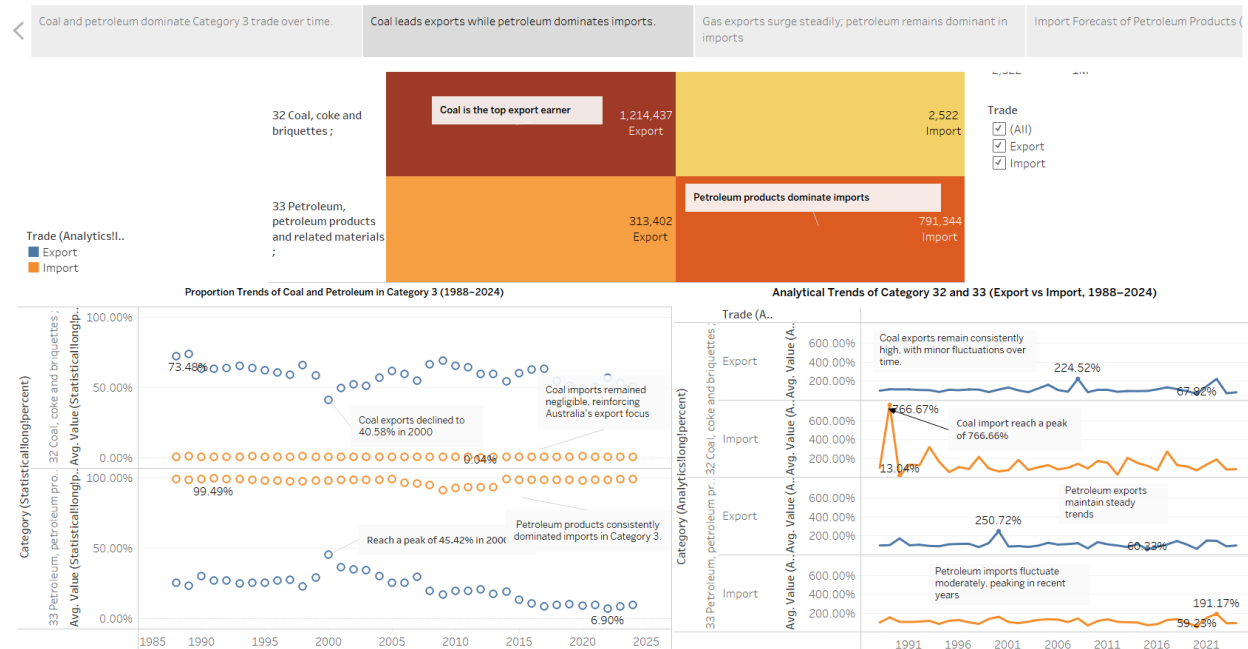
The first story highlights the evolving trade patterns of petroleum products within Category 3 from 1988 to 2024, with a comparative view of coal and gas. Coal (subcategory 32) consistently led exports, peaking at over \$140,000 million in 2022. This surge is attributed to the post-COVID global energy demand recovery (International Energy Agency, 2022). Meanwhile, petroleum exports (subcategory 33) were lower but experienced a notable rise in 2022, likely due to fuel supply disruptions and geopolitical instability (Statista, 2023).

In contrast, petroleum dominates imports, reaching over \$65,000 million in 2022, driven by price volatility and growing demand. Gas and coal imports remained negligible, reflecting Australia's robust domestic coal production and its role as a major LNG exporter.

Proportionally, coal accounted for nearly 59% of Category 3's exports, while petroleum made up over 97% of its imports, indicating a structural trade imbalance shaped by Australia's limited refining infrastructure and dependence on global petroleum markets.

3.3.2. Story 2: Coal leads exports while petroleum dominates imports

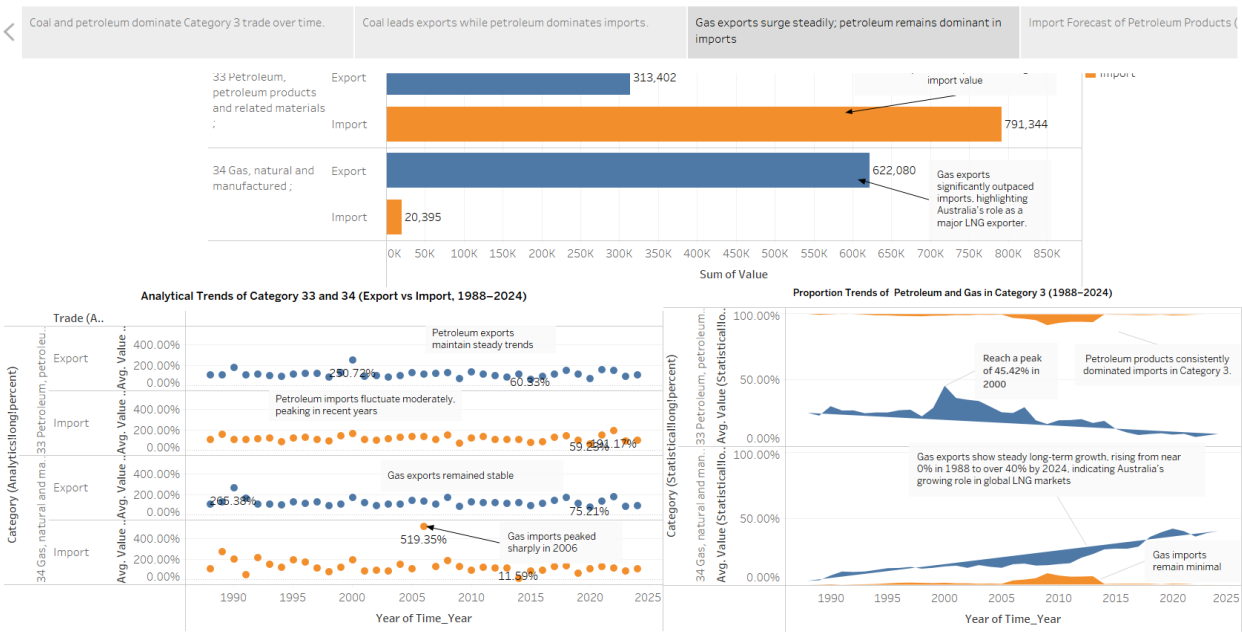
Energy Trade Focus: Petroleum Products (1988–2029)



This story highlights the contrasting trade roles of coal and petroleum in Category 3. Coal clearly dominates exports, while petroleum overwhelmingly leads imports, as shown by both total value and statistical proportions. Although coal exports remain consistently strong, petroleum imports fluctuate moderately and surged in recent years due to global price and supply chain pressures (Statista, 2023). The data reflects Australia's position as a major coal exporter and a petroleum-import-dependent economy.

3.3.3. Story 3: Gas exports surge steadily; petroleum remains dominant in imports

Energy Trade Focus: Petroleum Products (1988–2029)



Analytical Trends of Category 33 and 34 (Export vs Import, 1988–2024)

Proportion Trends of Petroleum and Gas in Category 3 (1988–2024)

Trade (A..

Category (Analytics)percent

Export

Import

33 Petroleum, petroleum products and related materials ;

Export

Import

34 Gas, natural and manufactured ;

Export

Import

Avg. Value (Statistical)percent

Year of Time_Year

Petroleum exports maintain steady trends

Petroleum imports fluctuate moderately, peaking in recent years

Gas exports remained stable

Gas imports peaked sharply in 2006

50.74%

60.53%

58.54%

75.21%

11.59%

519.35%

Category (Statistics)percent

33 Petroleum, petroleum products and related materials ;

34 Gas, natural and manufactured ;

Avg. Value (Statistical)percent

Year of Time_Year

Reach a peak of 45.42% in 2000

Petroleum products consistently dominated imports in Category 3.

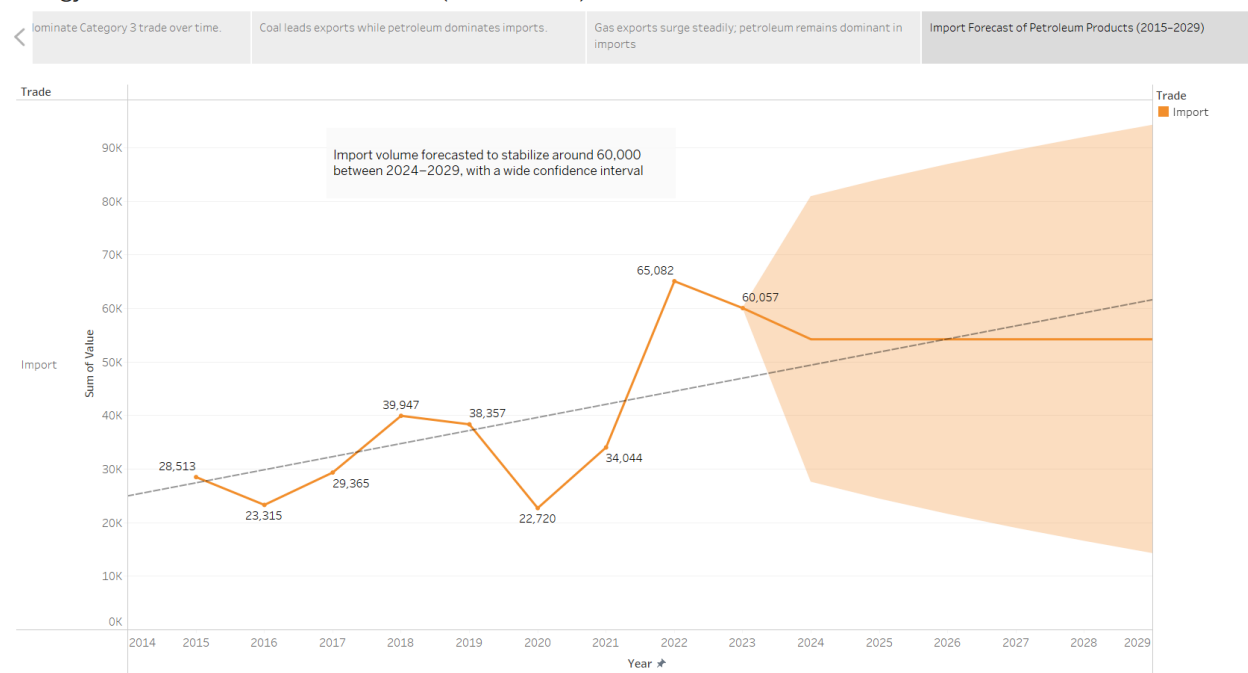
Gas exports show steady long-term growth, rising from near 0% in 1988 to over 40% by 2024, indicating Australia's growing role in global LNG markets

Gas imports remain minimal

This story compares the trade dynamics of petroleum and gas. While petroleum maintains dominance in imports with a high and steady trend, gas shows a different trajectory, exports have grown steadily, especially from the 2000s onward, highlighting Australia’s rise as a key LNG exporter (Department of Industry, Science and Resources, 2022). Despite a sharp import spike in 2006, gas imports remain minimal. The contrasting patterns reinforce Australia’s dual role: importer of refined petroleum and exporter of natural gas.

3.3.4. Story 4: Import Forecast of Petroleum Products (2015–2029)

Energy Trade Focus: Petroleum Products (1988–2029)



In this final story, we focus on forecasting petroleum product imports, the subcategory with the highest import value in Category 3. Historical data from 2015 to 2023 shows substantial volatility, especially a sharp surge in 2022 caused by global fuel price hikes and supply disruptions (IEA, 2022). Given its critical importance and consistent dominance in imports (over 97% of total Category 3 imports), petroleum was selected as the target product for import forecast.

Using historical data trends, the forecast suggests that petroleum import volumes will stabilize around \$60,000 million between 2024 and 2029, though with a wide confidence interval due to recent volatility. This insight supports the need for energy diversification or strategic stockpiling policies to mitigate future risks.

Between 2015 and 2024, petroleum product imports fluctuated significantly, peaking at \$65,082 million in 2022. Despite short-term gains from meeting domestic demand during global supply strain, this volatility posed a strategic risk due to price shocks and overreliance on external sources. The economic loss lies in increased import bills and exposure to geopolitical instability.

Looking forward from 2025 to 2029, the key target should be to stabilize import volumes around \$60,000 million, while investing in domestic refining capacity and diversifying

energy sources. This approach will reduce vulnerability and maintain secure energy supply under fluctuating global conditions.

IV. Recommendations and Methodological Reflection

The analysis of Category 3 trade (Coal, Petroleum, and Gas) from 1988 to 2029 reveals distinct patterns across subcategories, offering useful insights into national energy policy and trade strategy.

4.1 Recommendations

Coal remains Australia's leading export earner in Category 3, especially after peaking in 2022. However, future reliance on coal faces sustainability challenges due to global decarbonisation efforts. A gradual shift toward clean energy exports is advisable. Petroleum products dominate imports, often fluctuating in response to geopolitical and price shocks. Strengthening domestic refinery capabilities and diversifying sources could reduce vulnerability. Gas shows a steady export growth trend, positioning Australia as a major LNG supplier. It is recommended that Australia continue investing in LNG infrastructure and seek long-term contracts with key Asia-Pacific partners.

4.2 Methodological Reflection

4.2.1. Advantages

Dashboards offered a highly interactive and visually engaging platform that allowed users to explore complex data across multiple dimensions. By leveraging various chart types, such as line charts, treemaps, and stacked bar charts, dashboards enabled the identification of export-import dominance, trend volatility, and proportional trade contributions across subcategories. The ability to filter and drill down in real time further enhanced the user's capacity to examine data at both macro and micro levels, supporting detailed and responsive analysis.

Storyboards, in contrast, provided a more structured and narrative-driven approach. They facilitated the organization of insights into logical sequences, allowing key trends, such as the surge in coal exports or the petroleum import spike in 2022, to be communicated clearly and persuasively. The use of storyboards was particularly effective in highlighting breakthrough events and in constructing a strategic overview that aligned with the analytical findings from the dashboards.

4.2.2. Disadvantages

However, there were also several disadvantages associated with each method. While dashboards were powerful for exploratory analysis, they could become visually dense or difficult to interpret if not carefully designed, especially for users unfamiliar with the dataset or without explanatory annotations. On the other hand, storyboards were less flexible, as they followed a fixed sequence and did not support interactive filtering or real-time exploration. This made them less suitable for ad-hoc analysis or for adjusting perspectives based on evolving data needs. Additionally, if not clearly differentiated in purpose, dashboards and storyboards could contain overlapping content, which might result in redundancy or dilute the impact of the insights.

V. Summary and Conclusion

This study provides a comprehensive analysis of Category 3: Energy-related commodities (Coal, Petroleum, and Gas), focusing on their trade trends between 1988 and 2024, and forecasting to 2029. Through the four well-designed dashboards, the analysis captures critical trends, breakthrough shifts, and comparative dynamics among subcategories.

Coal emerged as the dominant export contributor, peaking in 2022 due to post-pandemic global demand (IEA, 2022). In contrast, petroleum products consistently dominated imports, reflecting Australia's dependency on foreign refined oil sources, particularly during global supply chain disruptions (Statista, 2023). Meanwhile, gas exports displayed a steady upward trend, affirming Australia's growing leadership in global LNG markets.

Breakthrough events such as the petroleum import peak in 2022 and coal's record export year offer key insights. Trend and forecast analysis for the next five years (2025–2029) suggests petroleum imports may stabilize around 60,000 units, albeit with uncertainty, and exports are projected to maintain moderate growth. Based on the 10-year trend and 5-year forecast, our recommendation is to strengthen refinery resilience, explore long-term LNG trade partnerships, and gradually reduce overreliance on petroleum imports.

The use of dashboards allowed efficient comparison of subcategories and timeframes, while the storyboard structured the narrative to clearly show evolution, dominance shifts, and future directions. Together, these tools enhanced interpretability and strategic insight, making them highly effective for industry and policy decision-making.

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